Editors' Suggestion

Hydrodynamic Spin-Coupling of Rotors

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Flow-mediated interactions between objects are important in sedimentation and collective locomotion. New phenomena arise in systems that are dominantly coupled by rotation, as relevant to active matter physics and biological and engineering contexts. Here we establish an experimental platform for studying fluid mechanical spin-coupling effects and map out how interactions depend on proximity, confinement, and flow state. The two-body problem of a driven rotor inducing spin of a passive rotor reveals a gearlike counterrotation mode but also surprising corotation modes that occur broadly across the parameter space. Streamline analysis and a mechanical model indicate distinct transitions between modes that arise geometrically at close range, topologically due to flow reconfiguration, and inertially with increasing Reynolds number. These findings provide ways to understand, control, and exploit hydrodynamic spin-spin interactions in applications.

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Flow-mediated interactions are well studied for bodies undergoing collective translation, as occurs in sedimentation, swarming, and flocking [1–8]. Recent studies in active matter physics have uncovered new phenomena that arise among objects whose rotation strongly modifies the nature of the hydrodynamic interactions [9,10]. Spinning biological cells and systems of actively driven colloidal particles exhibit collective orbiting, clustering into dynamic crystals, and edge currents, all of which have been attributed to viscous flow coupling [11–19]. Rotational interactions at larger scales are relevant to industrial applications, notably wind farms and turbine arrays where aerodynamic interference arises from swirl and inertial wake flows [20]. Recent work at intermediate scales and moderate Reynolds numbers has shown that spinning bodies cluster and collectively migrate, and flow modeling indicates that viscous and inertial effects are both essential [21]. Despite the prevalence and growing interest in spin-coupled systems, much of the rotor-scale fluid mechanics and multi-rotor coupling mechanisms remains poorly understood.

Idealized and minimal settings can serve as archetypes where the key effects are most cleanly revealed and studied. For the case of rotational interactions, classical examples involve vortices, pairs of which may orbit or propel as dipoles and collections of which may leap-frog, scatter, merge, and break down [22–26]. Spinning solid structures have been classically studied in the Taylor-Couette setting of concentric cylinders, where the emphasis is on the rich phenomenology and instabilities seen in the fluid that occupies the gap [27–30]. Similarly, multiple rotors have been studied within cylindrical corrals (i.e., vortex mixing flows) and in large periodic arrays (i.e., cellular flows), where again the rotations of solid cylinders are prescribed and the focus is on the mixing flows [31–39]. What is apparently missing is a system for the controlled study of the flow-structure interaction problem in which spin-coupling effects among multiple rotors can be revealed through their emergent motions.

Here we propose the two-rotor problem as an idealized setting for the study of rotational hydrodynamic interactions. As shown in Fig. 1(a), an active cylindrical rotor is driven with prescribed angular velocity about its axis, and a passive "test rotor" is placed nearby and may freely rotate in response to fluidic coupling. Confinement effects are explored through the size of the fluid domain or corral, while the strength and range of the interactions are assessed by changing the active speed and the interrotor separation distance.

Our realization as a lab experiment is shown schematically in Fig. 1(b), and additional details are available in Supplemental Material [40]. The corral and rotors are immersed in a large rectangular tank filled with a glycerol-water mixture whose viscosity is measured for each trial and falls in the range $\nu = 1.5$ to 2 cm^2/s. The rotors have identical radius $r = 1.9$ cm, and their height $h = 14.6$ cm nearly spans the tank and yields a high aspect ratio. The corral radius R is varied by employing cylinders of different sizes. These components along with the tank walls are made of clear acrylic plastic to allow for flow

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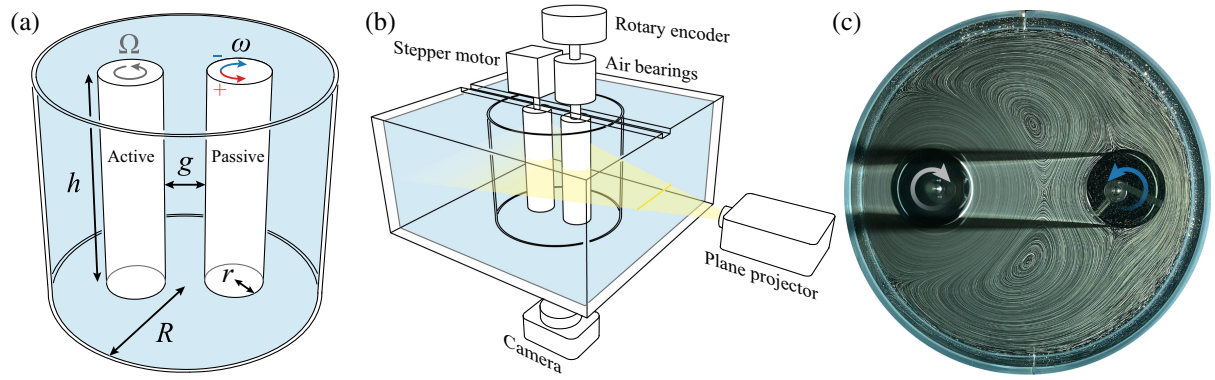


FIG. 1. Two-rotor problem. (a) An active cylinder driven to spin and which may induce spin of a passive cylinder due to hydrodynamic coupling. (b) Experimental apparatus (not to scale) showing actuation and suspension systems and means for measuring rotation and flow. (c) Example photograph of the counterrotation state (Re = 20 and $G = 0.6$).

imaging. The active rotor connects via a shaft to a programmable stepper motor that sets the angular velocity Ω, which will be systematically varied. The shaft to the passive rotor is suspended on low-friction air bearings, and a precision rotary encoder measures the angular position from which the average speed ω is derived. The rigid lid of the tank has slit openings that allow the separation gap g to be varied continuously over the entire allowable range thanks to a gantry (not shown) with translation stages. Flow visualization is achieved with a bottom-mounted camera and planar light sheet from a projector. The fluid is densely seeded with air bubbles whose typical size of 0.3 mm ensures minimal rise during time-exposed photographs, which capture flow path lines as shown in Fig. 1(c).

A dimensionless characterization involves three input parameters and one output quantity:

$$\mathrm{Re}=\frac{\Omega r^2}{\nu},\quad G=\frac{g}{2R-4r},\quad C=\frac{R}{r},\quad \Gamma=\frac{\omega}{\Omega}. \tag{1}$$

The Reynolds number Re involves the kinematic viscosity ν of the fluid and characterizes the actively driven flows. Intermediate values Re $\sim$ 10 to 100 are of particular interest as bridging the viscous and inertial regimes. The form of the dimensionless gap $G \in [0, 1]$ is chosen such that the extremes are set by rotor-rotor and rotor-corral contact. A range of corral sizes $C \in (2, 10)$ studied here permits assessment of finite-domain effects. Unlisted here among input parameters is the rotor aspect ratio $h/r = 8$, which is fixed and sufficiently large so that the couplings are dominantly determined by quasi-2D effects [40]. The dependent quantity is the speed ratio or "gear ratio" Γ that, as for mechanical gears, expresses the number of turns of the passive, output rotor relative to that of the active, input rotor. Its sign indicates corotation (+) or counterrotation (−).

Our experimental procedure involves fixing G and C and incrementally increasing Re by varying Ω in a step-wise fashion while continuously recording ω. Each step is sufficiently long so that the passive rotor reaches steady-state rotation [40]. The representative run displayed in Fig. 2(a) shows the staircase progression of the imposed active speed $\Omega(t)$ as the gray curve (right axis) and the corresponding response in the passive speed $\omega(t)$ as the blue-red data (left axis). Averaging $\omega(t)$ during the steady-state period within each step yields the gear ratio Γ as plotted in the inset, where the bars indicate standard deviations of the time-series data. For low spin rates Ω of the active rotor, the passive rotor spins in the opposite direction ($\omega < 0$ and $\Gamma < 0$, blue data). Such gearlike counterrotation might be expected based on reasoning about stronger shearing action on the inner portion of the passive rotor. At moderate Ω, we observe an unexpected transition to corotation ($\omega > 0$ and $\Gamma > 0$, red data) that then persists for yet higher driving rates. Additional runs for decreasing Ω show no discernible hysteresis or bistability, indicating that the spin state is unique for given conditions.

Corotation states arise broadly across the parameter space, as revealed by sweeping Ω for values of $G \in (0, 1)$. The resulting phase diagram of Fig. 2(b) maps out the measured gear ratio Γ as a function of the input parameters (G, Re), where a total of 290 conditions are investigated. Again, counterrotation modes are in blue and corotation modes are in red, and the marked vertical transect corresponds to the conditions of Fig. 2(a). The map is complex yet structured with the modes situated in distinct regions separated by stagnation boundaries (white) at which the passive rotor does not spin. Other notable aspects include the fine structure and sensitive response for low G and the nonmonotonic variations of the coupling strength $|\Gamma|$ across the whole phase space, including tongue-shaped regions of counterrotation that extend into high Re.

We further explore the three-dimensional space of the input parameters by assessing the corral size C, as documented in Fig. 2(c). Here the planar slice marked by an arrow corresponds to the conditions of Fig. 2(b), and

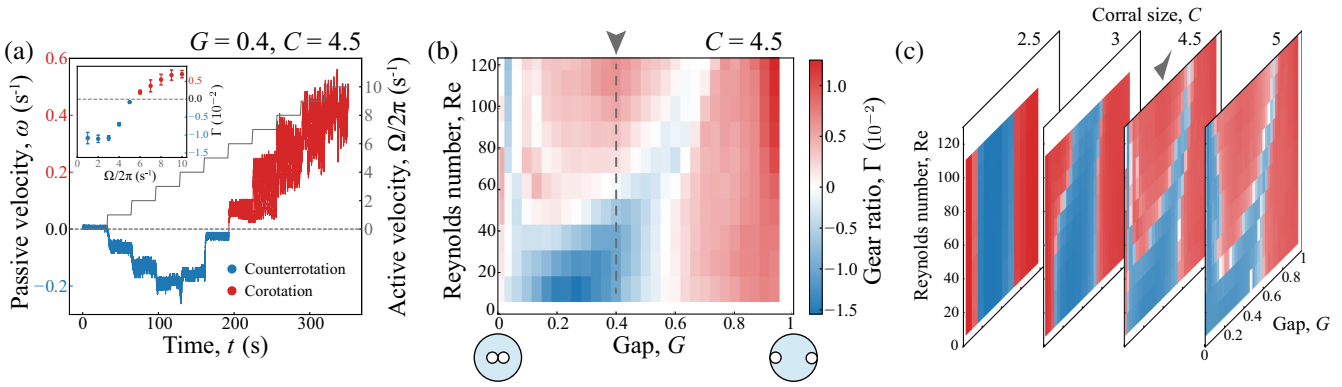


FIG. 2. System response across parameters. (a) Experimental procedure and data: the active angular velocity (gray curve and right axis) is increased in steps long enough to allow for equilibration of the passive velocity (blue-red curve and left axis). Inset: passive-to-active speed or gear ratio shown as the mean and standard deviation measured during each step. (b) Map of the gear ratio for varying the interrotor gap and driving Reynolds numbers $\mathrm{Re} \propto \Omega$ at fixed corral size. The indicated vertical transect corresponds to the data of (a). (c) Assessment of different corral sizes C, colored logarithmically.

similar sweeps of (G, Re) are conducted for each C to yield a grand total of 1,350 conditions. Taken together, these data further confirm that corotation states (red) occur broadly, and they show that counter-to-corotation transitions can be achieved by varying any of the three input parameters. Many aspects defy intuition. For example, corotation often prevails at the smallest G, where close proximity would suggest that the rotors may spin oppositely like a pair of gears. Corotation also occurs in all cases for higher gaps, and the increase of coupling strength $|\Gamma|$ with G contradicts the expectation that hydrodynamic interactions decay with distance.

We seek to explain these observations by assessing the flow fields visualized by time-exposed photographs and Supplemental Material videos available online. For this purpose, we focus on the case of $C = 4.5$ whose results are mapped out in Fig. 2(b) and which display the full complexity of possible outcomes in terms of spin states. Its structure suggests convenient transects through the space of (G, Re), along which all the key states are observed. A horizontal transect at fixed $\mathrm{Re} = 20$ and increasing G reveals a reentrant transition: corotation gives way to counterrotation and then back to corotation. A vertical transect at, e.g., fixed $G = 0.3$ reveals a single counter-to-corotation transition for increasing Re. These two pathways are assessed in Figs. 3(a) and 3(c) respectively via path line photographs. The cartoon phase map of Fig. 4(a) provides helpful orientation relating the pathways, states, and transitions.

The active rotor swirls up an extended flow circulation that varies with (G, Re). The passive rotor is immersed in the wash of this flow that, upon close inspection, is seen to induce a stagnation point on the windward side of the cylinder and a separation point on the downstream side. These points are most clearly seen in the enlarged images of Fig. 3(c), and they divide the circumferential surface of the passive rotor into two parts: an inner portion closer to the active rotor and along which the shear is signed so as to lead to torque of one sense and an outer portion whose shear-induced torque is oppositely directed. These observations suggest a picture of competing shear forces, the balance of which determines the sense of rotation. This view is supported by estimates of the shear stress on the passive rotor due to boundary layer flows induced by the active rotor, which are shown to account for the scale of Γ [40].

The flows for the case of varying the interrotor gap distance are documented in Fig. 3(a). The middle image of (a) represents a base case at moderate G for which gearlike counterrotation is intuitively explained as arising from stronger shear on the inside. The left image represents rotors brought very close together, which pinches down the inner shear zone and apparently tips the balance in favor of the outer shear whose direction promotes corotation. The right image represents the case of wide separations, which exhibit corotation by an entirely different mechanism. The flow field is dramatically altered by an intervening recirculation that emerges between the rotors, a feature noted in numerical simulations [19]. The washing flow in the vicinity of the passive rotor is now reversed, and the inner portion maintains dominance over the outer but with the reverse direction compared to the base case. A survey of the flow fields as a function of G reveals a sequence of topological transitions [40].

The flows for the case of varying Reynolds numbers are documented in Fig. 3(c). The single transition can be understood by comparing the base case of counterrotation (leftmost images) and the corotation state (rightmost images). Here the washing flow is downwardly directed in both, and the relevant changes involve shrinkage of the inner shear zone with increasing Re as well as enhanced flow speed on the outer side of the rotor. A candidate explanation consistent with inertial flows is centrifugal action [41,42], which causes the flow spun up around the active rotor to increasingly fling outward at higher Re,

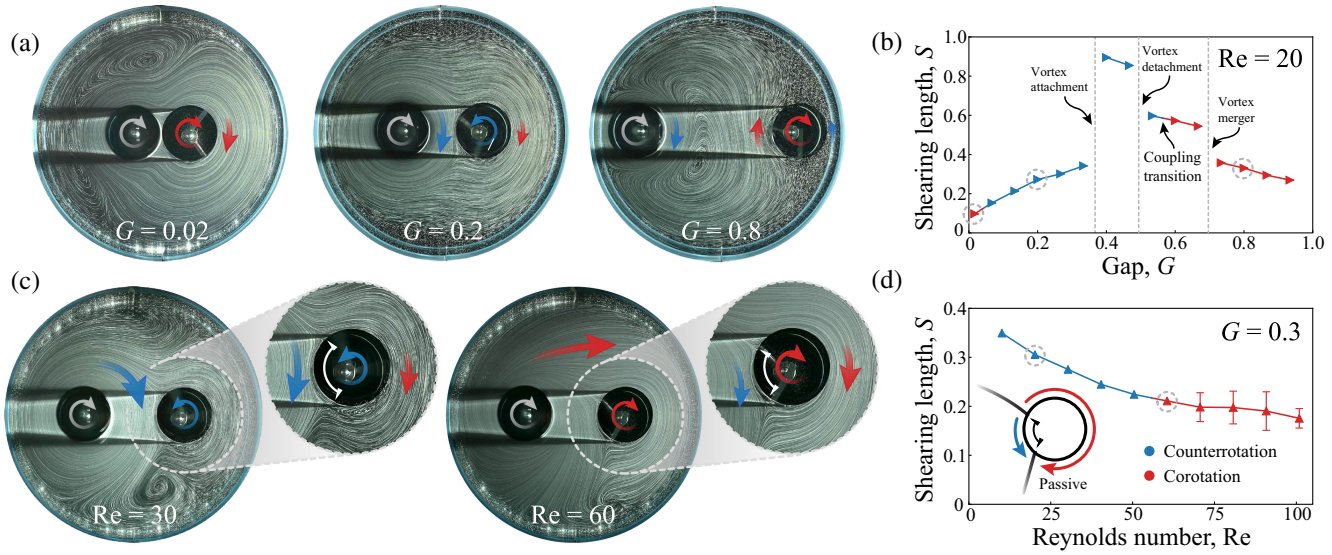


FIG. 3. Flow-induced coupling transitions and shear competition. (a) Spin reversals of the passive rotor induced by the increasing gap at Re = 20. The inner shear dominates the opposing outer shear at both $G = 0.2$ and 0.8, but the shears reverse sign due to global flow reconfiguration. (b) Length of inner shear zone along the circumference of the passive rotor. Flow rearrangements enable, but do not coincide with, coupling transitions for varying the gap. The conditions of (a) are circled. (c) Spin reversal induced at higher Re for a fixed $G = 0.3$. The inner shear zone between the stagnation and separation points shrinks with increasing Re, and the flow around the active rotor transitions from closed loops to outward spirals. (d) Shrinkage of the inner shear zone with Re. The conditions of (c) are circled, and vertical bars represent variations due to flow unsteadiness.

leading to the outward spiraling path lines seen in Fig. 3(c). An empirical model of the shear informed by the flow observations quantitatively reproduces the critical value $\mathrm{Re} \approx 55$ marking the transition [40].

The global and local changes in flows are reflected in the inner shearing length $S \in [0, 1]$, defined as the arc length between the stagnation and separation points along which shearing promotes counterrotation and normalized by the circumference $2\pi r$ [see the inset of Fig. 3(d)]. Figures 3(b) and 3(d) chart the measured values along horizontal and vertical pathways through the space of (G, Re), and the data markers are colored according to counterrotation (blue) or corotation (red). The corotation states at very small G in (b) and at high Re in (d) can be understood as arising from shrinking S that allows the outer shear to prevail. By contrast, the cases at higher G in (b) are fundamentally different due to global flow rearrangements, of which there are several and each of which is marked by an abrupt change in S. The rearrangements involve various events such as detachment of a vortex from the corral wall, attachment to and detachment from the passive rotor, and merging of vortices [40]. Surprisingly, none of these events coincide exactly with the spin transitions, but rather their net effect enables the transition due to reversal of the flow washing the passive rotor. The events show no discernible hysteresis [40], implying uniqueness of the flow state.

A mechanical analogy helps to reconcile how competing inner and outer shear forces determine spin states and their transitions. The picture given in Fig. 4(b) is that of two rotors, the left undergoing prescribed rotation and the right responding through two coupling mechanisms: toothed gears engage on their inner portions to promote counterrotation and a belt engages on their outer portions to promote corotation. Figure 4(c) illustrates four reference cases, including the base case of counterrotation (bottom middle) and three corotation states at the right, left, and top. These are laid out to parallel the phase diagram of (a) where the states are marked 0-3. For the base case marked 0 in (a) and corresponding to moderate G and low Re, the dominance of the inner shear over the outer leads to counterrotation, for which the gears engage and the belt slips. For the corotation mode marked 1 with the rotors widely

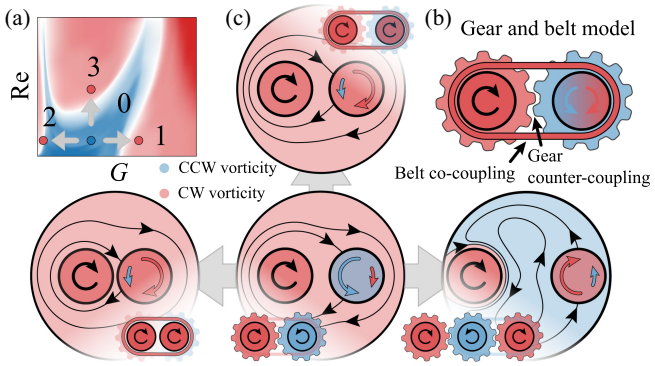


FIG. 4. Gear versus belt coupling. (a) Cartoon phase diagram showing three routes from counterrotation (blue) to corotation (red). (b) Mechanical analogy of gearlike inner coupling versus beltlike outer coupling. (c) Streamlines, shear competitions, and mechanical interpretations of transitions.

separated, the inner shear again dominates but reverses direction, which corresponds to the insertion of a middle gear in a train. For the corotation mode marked 2 with the rotors pressed tightly together, the gears grind and lose purchase and the belt wins out. For corotation mode 3 at higher Re, the centrifugally enhanced outer looping flow corresponds to strengthening of the belt mode.

The analogy helps interpret details of the phase map such as the tonguelike regions of counterrotation that separate the three corotation modes at higher Re. Noting the tableaux of Fig. 2(c), small corral size C is structured such that corotation modes 1 and 2 are on either side of the counterrotation mode 0. For larger C, mode 3 enters the phase diagram by invading mode 0 from above (high Re), and the tongues are the result of mode 0 being cleaved down the middle. This confirms that the three corotation modes are indeed distinct, with the tongues representing phase boundaries between pairs of such modes. For example, the right tongue represents the conflict between the belt mode 3 on one side and the gear train mode 1 on the other, both of which promote corotation but through wholly distinct mechanisms that involve opposite flow and shear directions.

These findings reveal a complex but structured phase diagram for hydrodynamic spin coupling. The expected gearlike counterrotation of a passive rotor is accompanied by beltlike corotation states that prove to be as prominent across the parameter space of interrotor gap, Reynolds number, and corral size. Three distinct counter-to-corotation pathways are identified, and the causal factors in all cases involve shear competitions. A geometric flow transition occurs when the rotors are in very close proximity, the inner shear zone is contracted, and the inner flow constricted. By contrast, a topological transition occurs at wider gaps and is marked by global rearrangements of the intervening flow that reverses the inner shear. An inertial transition occurs at higher Re, where the observed strengthening of the outer flows may be due to centrifugal action. While they may manifest differently in other systems, these effects are likely to play similarly important roles in the application areas that motivate this study.

There remains much to understand about hydrodynamic spin coupling. Realizing larger ensembles would permit the study of how the two-body interactions outlined here relate to and perhaps explain collective spin states [21]. Future studies might consider varying the rotor and corral geometries, permitting free translation of the rotors [11,12], or deploying flow-based measurements of torque [43,44]. Such pursuits would benefit from numerical simulations and flow-structure interaction models [45,46], for which the archetypal system and its characterization presented here serve as benchmarks.

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Data availability—The data are not publicly available. The data are available from the authors upon reasonable request.

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